

Project Report 1

Goal of this week:

To define the project structure clearly and decide the correct development order for the main modules: PID/LQR, PPO/SAC, and CVAE/VAE.

- **PID (Proportional–Integral–Derivative):** A classical feedback control method that reduces the error between a desired target and the actual system output using proportional, integral, and derivative terms.
- **LQR (Linear Quadratic Regulator):** An optimal control method that computes control inputs by minimizing a cost function that balances tracking performance and control effort.
- **PPO (Proximal Policy Optimization):** A reinforcement learning algorithm that improves a policy in a stable way by limiting how much the policy can change at each update step.
- **SAC (Soft Actor-Critic):** A reinforcement learning algorithm for continuous control that learns efficiently by maximizing both reward and policy entropy, encouraging exploration.
- **CVAE (Conditional Variational Autoencoder):** A generative model that learns to produce data conditioned on additional information, such as labels or context variables.
- **VAE (Variational Autoencoder):** A generative model that learns a compact latent representation of data and can generate new similar samples by decoding points from that latent space.

What I achieved:

I organized the overall architecture of the project around a 2D physics environment, drone module, and wind simulation module. I also clarified that the most critical parts of the project are the control, RL, and generative AI components.

What I could not complete and why:

I planned to begin with the physics engine, but I could not complete it this week because it is not a simple preliminary step. It is a major and time-consuming part of the project that requires substantial implementation and testing before the AI modules can be integrated.

Goal for next week:

To implement a VAE using the MNIST dataset, learn the generative modeling pipeline in practice, and use this as preparation for future CVAE-based wind generation.